# MUHAMMAD HASAN FAROUK

Cairo, Egypt (+20) 115 852 2627 Email Website
LinkedIn GitHub LeetCode

## Professional Summary

Final-year Computer Science student specializing in **High-Performance Computing** and **Scalable Backend Engineering**. Proven expertise in architecting computational infrastructure, notably leading the end-to-end reconstruction of the FCAI-CU R&D Lab's Beowulf cluster. Recently certified in Efficient Scientific Computing by **INFN-SESAME**, with a deep technical focus on **parallel programming MPI and OpenMP**, **GPU acceleration CUDA**, and heterogeneous systems. Adept at bridging the gap between low-level hardware optimization and high-level backend architecture to deliver performance-critical solutions.

## Education

**Cairo University** — Cairo, Egypt
B.Sc. in Computer Science — Expected 2026
GPA: 3.27 / 4.0

## Technical Skills

- **Programming Languages:** C++, C, Java, Python, C#, JavaScript, SQL, Bash.
- **Parallel & High-Performance Computing:** MPI, CUDA, OpenMP, Intel TBB, Multithreading.
- **Backend Development:** Spring Boot, Django, REST APIs, JWT Authentication, PostgreSQL.
- **DevOps & Systems:** Docker, Kubernetes, Linux System Administration, SSH, Git, GitHub.

## HPC & Systems Engineering Experience

**HPC R&D Intern** — July 2025 – Sept 2025
*Faculty of Computers and Artificial Intelligence, Cairo University* — Cairo, Egypt

- **HiPer-FC Beowulf Cluster:** Designed and implemented a distributed computing system built entirely from commodity hardware, translating theoretical HPC architecture into real-world deployment.
- **System Engineering:** Integrated isolated workstations into a unified cluster via a high-speed network backbone; deployed **Rocky Linux** and configured secure SSH-based automated node management.
- **Parallel Execution:** Configured MPI and OpenMP environments, enabling scalable parallel execution of research workloads across multiple nodes.
- **Scaling Initiative:** Collaborated with the R&D team to plan migration toward enterprise-grade servers, establishing a centralized HPC hub to expand student research accessibility.

## International Training & Certifications

**International School on Efficient Scientific Computing** — Feb 1–6, 2026
*INFN & SESAME* — Allan, Jordan

- **Distributed & Shared Parallelism:** Implemented MPI-based distributed applications and task-parallel workloads using Intel TBB and work-stealing schedulers.
- **GPU Acceleration:** Developed optimized CUDA kernels for heterogeneous computing, focusing on thread hierarchy optimization and memory latency reduction.
- **Modern C++ & Memory:** Applied RAII, move semantics, and smart pointers; analyzed cache coherence effects and mitigated **false sharing**.

## Web Development Projects

**Learning Management System (LMS)** — *Spring Boot, PostgreSQL, JWT*

- Developed a secure RESTful backend for managing online courses, implementing **JWT**-based stateless authentication.
- Implemented application-level caching using Spring Boot's `@Cacheable`, significantly reducing database load and improving response latency.
- Integrated SMTP-based automated email notifications for system updates and course events.

**Restaurant Management System** — *C#, Entity Framework Core*

- Built a desktop-based order management solution using Entity Framework Core for ORM and schema migrations.
- Designed a clean layered architecture separating business logic from database persistence.

**Web Book Shop** — *Django, HTML/CSS*

- Developed a full MVC-based e-commerce platform with secure authentication and relational data modeling using Django ORM.